SCHOOL OF ENGINEERING,CUSAT

B. TECH DEGREE 2nd SEMESTER FIRST INTERNAL EXAMINATIONS, APRIL 2024

23-200-0205B INTRODUCTION TO CYBER PHYSICAL SYSTEMS

(2023 Scheme)

TIME: 2 hrs **MAX. MARKS: 30**

Course Outcomes

On completion of this course the student will be able to:

CO1	Understand the features & components of Cyber Physical Systems
CO2	Understand the elementary constructs of Arduino Software
CO3	Develop optimal programs & circuits for interfacing various sensors with Arduino
CO4	Apply Arduino IDE for developing suitable programs for interfacing sensors & actuators with Arduino & Node MCU

PART A

(Answer all questions) **(5 x 2= 10 Marks)**

	Questions	Marks	BL	CO	PO
I (a)	What are the basic data types used in Arduino programming?	2	L1	2	1
(b)	Briefly explain the concept of sensors and actuators in the context of IoT.	2	L2	1	3
(c)	Describe the role of the Arduino IDE.	2	L2	2	1
(d)	What is Cyber-Physical Systems? List out the applications of CPS.	2	L3	1	2
(e)	Define operators, functions and arrays.	2	L4	2	1

PART B

(Answer the following questions choosing either the first OR second option)

(1 x 10 = 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Explain the general structure of CPS.	5	L1	1	3
(b)	Explain IOT stack and levels of IOT.	5	L1	1	2
	OR				
III (a)	What are the key features of CPS, and how do they contribute to the system's functionality?	6	L2	1	1
(b)	Describe the time functions available in Arduino.	4	L1	2	1
	(1 x 10 = 10 Marks)				
IV (a)	What are the main differences between Arduino Uno and Node MCU development boards in terms of features, capabilities, and target applications?	4	L2	2	1
(b)	Discuss the enabling technologies that support the development and deployment of IoT systems.	6	L1	1	3
	OR				
V (a)	Explain the general structure of an Arduino Program with an example.	4	L4	1	1
(b)	What are the characteristics of the Internet of Things (IoT), and how do they contribute to its widespread adoption in various industries?	6	L2	1	2

Bloom's Taxonomy Levels

L1-%	L2- %	L3-%	L4-%	L5-%	L6-%
38	38	8	16	0	0